AP CALCULUS AB

Unit 1 • Lesson 1-2 • Guided notes

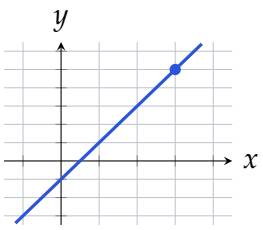
Lesson 1-2 • Limits to a Point of a Function

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: evaluate limits algebraically, including when plugging in fails.

Evaluating limits

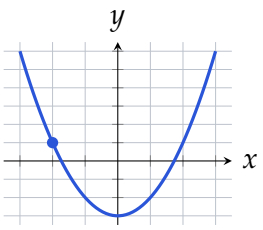
Linear $f(x) = 2x - 1$



$\lim_{x \to 3} f(x) =$ ____

$f(3) =$ ____

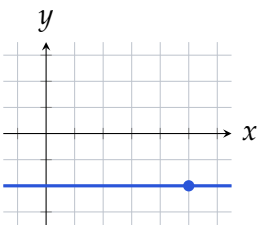
Quadratic $g(x) = x^2 - 3$



$\lim_{x \to -2} g(x) =$ ____

$g(-2) =$ ____

Constant $h(x) = -2$



$\lim_{x \to 5} h(x) =$ _____

$h(5) =$ _____

For polynomials (and constants), the limit equals the value: just _____ c.

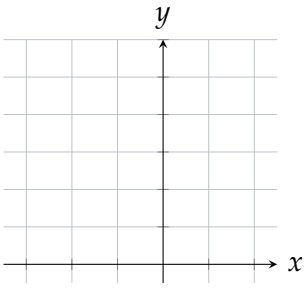
Limits of piecewise functions

$$j(x) = \begin{cases} x + 3, & x \leq 1 \\ x^2 + 1, & x > 1 \end{cases}$$

$x = 1$ is the _____: use the _____ piece for the left and the _____ piece for the right.

$\lim_{x \to 1^-} j(x) =$ ____ $\lim_{x \to 1^+} j(x) =$ ____

$\lim_{x \to 1} j(x) =$ _____ $j(1) =$ ____



When plugging in fails

$\lim_{x \to 3} \dfrac{x^2 - 9}{x^2 - x - 6}$ Plug in: ____, the _____. It means: do more work!

Three steps:

1) _____

2) _____

3) _____

Example 1	$\lim_{x \to 0} \frac{x^2 - 4x}{3x^2 + x}$
Example 2	$\lim_{x \to -3} \frac{x^2 + 6x + 9}{x + 3}$
Summary	